TERNARY CODES WITH SYMMETRY AND HOLOMORPHIC SCFTS OF MINIMAL INDEX

[DRAFT — AUTHORS TO BE ADDED]

Abstract. Gaiotto and Johnson-Freyd asked whether for every k there is a holomorphic $\mathcal{N}=1$ vertex operator superalgebra of central charge $12k$ whose supersymmetric index is $24/\gcd(k,24)$, the smallest value allowed by topological modular forms. They answered this for $k \le 5$ by orbifolding the SVOA of a self-dual ternary code of index 24 by a code automorphism of order $\gcd(k,24)$ with prescribed properties, the codes being found by computer search. We make the partial-quadrupling construction of index-24 codes equivariant: it commutes with code automorphisms that permute its blocks, and the twined index of an order-3 automorphism is carried along as a block-weighted index of a collapsed fixed code. Starting from the Golay code and from one computer-found code of length 36 with a symmetry of order 12, we obtain for every k not divisible by 8 an explicit code and automorphism satisfying their hypotheses. This answers the question for all $k \not\equiv 0 \pmod 8$. Conversely, we show that a self-dual ternary code with an automorphism g satisfying $g^4 = -1$ has index divisible by 48, so the same cyclic-orbifold method cannot reach $k \equiv 0 \pmod 8$. We then compute the 't Hooft anomaly of a quaternion group Q_8 of code automorphisms: by a reduction to free fermions it is $3k$ times a class of order 8, so it vanishes exactly when $8 \mid k$. Orbifolding by Q_8 answers the question for $k \equiv 8, 16 \pmod{24}$. Finally we describe Q_8-invariant codes as $\mathbb{F}_3^2 \otimes U$ with U Lagrangian, which makes their index computable at length 72. An obstruction seen at length 36 disappears there, and a seed found at length 72 gives a $Q_8 \times \mathbb{Z}/3$-equivariant chain whose orbifolds have index 1 for $k \equiv 0 \pmod{24}$. So the Main Question has a positive answer for every k.

1. Introduction

A self-dual ternary code $C \subseteq \mathbb{F}_3^n$ determines a holomorphic $\mathcal{N}=1$ SVOA $V_{\Lambda(C)}$ of central charge n [6]. Its supersymmetric (Ramond–Ramond) index is

$$\mathrm{Index}(C) = \#\{\text{even maximal codewords}\} - \#\{\text{odd maximal codewords}\},$$

where a codeword is maximal if it has no zero entry and its parity is that of the number of entries equal to 1. The index is defined up to sign (negating a coordinate negates it), and is 0 unless $12 \mid n$. Put $\varphi(0)=0$, $\varphi(1)=-1$, $\varphi(2)=1$ and $f(x)=\prod_i \varphi(x_i)$, so that $\mathrm{Index}(C)=\sum_{c\in C} f(c)$; for any subset $X \subseteq \mathbb{F}_3^n$ write $F(X)=\sum_{x\in X} f(x)$.

By the Stolz–Teichner programme and the computation of the image of $\pi_{24k}\mathrm{TMF} \to \mathrm{MF}_{12k}$ [7], the index of a holomorphic $\mathcal{N}=1$ SCFT of central charge $12k$ should be divisible by $24/\gcd(k,24)$. Gaiotto and Johnson-Freyd ask for the converse:

Main Question ([6])**.** *For each k, does there exist a holomorphic $\mathcal{N}=1$ VOA with central charge $12k$ and supersymmetric index $24/\gcd(k,24)$?*

Their tool is the following theorem. The automorphism group $\mathrm{Aut}(C)$ is taken inside the group $2^n{:}S_n$ of signed coordinate permutations; it always contains -1.

Theorem 1.1 ([6, Thm. 5.6])**.** *Let C be self-dual of length $12k$ with $\mathrm{Index}(C)=24$, put $m=\gcd(24,k)$, and let $g \in \mathrm{Aut}(C)$ have order m. Suppose that (1) if m is even then $g^{m/2}=-1$; and (2) if $3 \mid m$ then $g^{m/3}$ fixes as many even as odd maximal codewords and permutes the*

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coordinates in 3-cycles only. Then g lifts to a non-anomalous automorphism of $V_{\Lambda(C)}$ and the orbifold $V_{\Lambda(C)}/\!\!/\langle g\rangle$ has index $24/m$.

Negating one coordinate of C and conjugating g by that sign change preserves the hypotheses on g (the parity of every maximal codeword flips at once) and negates the index, so $\mathrm{Index}(C) = \pm 24$ suffices. With codes found by computer search, this answers the Main Question for $k \le 5$ [6, Ex. 5.7]. In [2] we constructed self-dual codes of index 24 in every length $12k$ by a *partial quadrupling* Q_σ (recalled in §2). Here we show how to build the required automorphism into that construction.

Theorem A. *For every $k \ge 1$ with $8 \nmid k$ there is an explicit self-dual ternary code C of length $12k$ with $\mathrm{Index}(C) = \pm 24$ and an automorphism g satisfying the hypotheses of Theorem 1.1. Hence the Main Question has a positive answer for all $k \not\equiv 0 \pmod 8$.*

The residue classes $\gcd(k, 24) \in \{1, 2\}$ need nothing new (§4.1); $\gcd(k, 24) = 4$ uses an automorphism J of the Golay code with $J^2 = -1$ (§4.2); $\gcd(k, 24) \in \{3, 6, 12\}$ uses one code of length 36 with an automorphism of order 12, found by the method of [6] (§4.3). The remaining classes $k \equiv 0, 8, 16 \pmod{24}$ are out of reach of Theorem 1.1:

Theorem B. *Let C be a self-dual ternary code of length $12k$ with an automorphism g such that $g^4 = -1$. Then $48 \mid \mathrm{Index}(C)$, and $\mathrm{Index}(C) = 0$ if k is odd. In particular, when $8 \mid k$ no code of index 24 admits an automorphism as in Theorem 1.1.*

Indeed for $m \in \{8, 24\}$ condition (1) asks for $g^{m/2} = -1$, and then $h = g^{m/8}$ satisfies $h^4 = -1$. For $k \equiv 8, 16 \pmod{24}$ a non-cyclic group does the job instead:

Theorem C. *If $Q_8 = \langle J, K\rangle \subseteq \mathrm{Aut}(C)$ with $J^2 = K^2 = -1$, the Q_8-action on $V_{\Lambda(C)}$ is non-anomalous if and only if $8 \mid k$. Codes of index ± 24 with such a Q_8 exist for every $k \equiv 0 \pmod 4$, so the Main Question also has a positive answer for $k \equiv 8, 16 \pmod{24}$.*

For $k \equiv 0 \pmod{24}$ we add an order-3 element to Q_8. At length 36 a vanishing twined count forces the index to be divisible by 72, but this is special to length 36. At length 72 a computer search, made feasible by describing Q_8-invariant codes as $\mathbb{F}_3^2 \otimes U$ with U Lagrangian (§6.1), finds a suitable seed:

Theorem D. *For every $k \equiv 0 \pmod{12}$, $k \ge 12$, there is an explicit self-dual ternary code of length $12k$ and index 24 with $G = Q_8 \times \mathbb{Z}/3 \subseteq \mathrm{Aut}(C)$, where $J^2 = K^2 = -1$ and the order-3 element h is fixed-point-free with vanishing twined count. For $k \equiv 0 \pmod{24}$ the orbifold $V_{\Lambda(C)}/\!\!/G$ has index 1. Hence the Main Question has a positive answer for every k.*

2. Partial quadrupling

Let $T \subseteq \mathbb{F}_3^4$ be the tetracode and $\lambda(0, 1, 2, 3, 4) = (0, -2, 0, 0, 1)$. A *block system* σ on a coordinate set X is a set of disjoint 4-subsets (blocks), each split into two pairs; the other coordinates are singletons. For a code $E \subseteq \mathbb{F}_3^X$ put $R_\sigma(E) = \sum_{e \in E} \prod_{i \text{ singleton}} \varphi(e_i) \prod_{\beta \in \sigma} \lambda(\mathrm{wt}(e|_\beta))$. For E self-dual of length m and $|\sigma| = b$, the code $Q_\sigma(E) \subseteq \mathbb{F}_3^{m+12b}$ is defined in [2, Def. 6]: a block $\beta = \{p, q\} \sqcup \{r, s\}$ receives new coordinates β^* carrying a copy of T, each pair (i, i^*) is glued by $h(x, y) = (x + y, x - y)$, the *step-one blocks* $\{p, p^*, q, q^*\}$ and $\{r, r^*, s, s^*\}$ receive new coordinates ** carrying copies of T, and each pair (y, y^{**}) is glued by h again. $Q_\sigma(E)$ is self-dual, and for every set $\mathcal{B}$ of step-one blocks, each split into two pairs $\{y_1, y_2\} \sqcup \{y_3, y_4\}$ and refined into $\{y_1, y_1^{**}, y_2, y_2^{**}\}$ and $\{y_3, y_3^{**}, y_4, y_4^{**}\}$, the resulting block system σ'' satisfies

$$\mathrm{Index}(Q_\sigma(E)) = R_\sigma(E), \qquad R_{\sigma''}(Q_\sigma(E)) = R_\sigma(E) \tag{1}$$

[2, Lemma 8]. The proof of (1) sums over the tetracode variables block by block, so both identities hold with Index replaced by F for *any* linear code E, self-dual or not.

3. Equivariance

Lemma 3.1. *Let $G \subseteq \mathrm{Aut}(E)$ be a finite group that permutes the blocks of σ and maps split pairs to split pairs. Extend each $g \in G$ to the new coordinates by letting i^* follow i with the same sign: if $g(e_i) = \pm e_j$ then $g(e_{i^*}) = \pm e_{j^*}$, and likewise for **. Choose the tetracodes G-equivariantly: on one block of each G-orbit a copy of T invariant under the stabiliser, and its transports elsewhere. Then $G \subseteq \mathrm{Aut}(Q_\sigma(E))$; the extension preserves orders, relations $g^j = -1$, and cycle lengths; and if $\mathcal{B}$ and its splits are G-stable, then σ'' is G-stable. The construction iterates provided stabiliser-invariant placements of T exist again at both steps of the next stage; this holds for all chains below when the splits are chosen so that the stabilisers fix both pairs (for a J-stable step-one block $\{p, p^*, q, q^*\}$ the split $\{p, q\} \sqcup \{p^*, q^*\}$, and for the resulting blocks $\{x, y, x^{**}, y^{**}\}$ the split $\{x, y\} \sqcup \{x^{**}, y^{**}\}$).*

Proof. On a glued pair (i, i^*) the extended g acts by a common sign, which commutes with h. So the two gluing maps commute with G, and $Q_\sigma(E)$ is the image of the G-invariant code $E \oplus T^b \oplus T^{2b}$. The remaining statements follow from the definition of the extension. □

A stabiliser-invariant copy of T exists in the cases we need. If J swaps $p \leftrightarrow q$ and $r \leftrightarrow s$ with $J^2 = -1$, say $J(e_0) = e_1$, $J(e_1) = -e_0$, $J(e_2) = e_3$, $J(e_3) = -e_2$, then the tetracode spanned by $(1, 1, 1, 0)$ and $(0, 1, 2, 2)$ is J-invariant. An element of order 3 without fixed coordinates stabilises no block, since $3 \nmid 4$. In particular, in the presence of such an element all block orbits have size divisible by 3, so $b \equiv 0 \pmod 3$ and $k \bmod 3$ is unchanged along the construction.

Lemma 3.2. *Let $h \in G$ have order 3 and no fixed coordinate, and let $T_h(C) = \sum_{x \in C,\ hx = x} f(x)$. Collapsing h-fixed vectors to one coordinate per h-orbit, let D be the collapsed fixed code of E and $\bar{\sigma}$ the collapsed block system (one block per h-orbit of blocks). Then*

$$T_h(Q_\sigma(E)) = R_{\bar{\sigma}}(D), \qquad T_h(Q_{\sigma''}(Q_\sigma(E))) = R_{\bar{\sigma}}(D) \quad \text{for } G\text{-stable } \sigma''.$$

Proof. After a diagonal sign change, which multiplies Index and T_h by the same sign, h permutes coordinates. An h-fixed vector is constant on h-orbits, and $\varphi^3 = \varphi$, so collapsing preserves f. Since $Q_\sigma(E)$ is the G-equivariant image of $E \oplus T^b \oplus T^{2b}$, its fixed vectors are the images of fixed vectors; the fixed vectors of an h-transported triple of tetracodes form one tetracode after collapsing, and the gluing maps commute with collapsing. Hence the collapsed fixed code of $Q_\sigma(E)$ is $Q_{\bar{\sigma}}(D)$, and (1) for the (possibly non-self-dual) code D gives both identities. □

4. Proof of Theorem A

4.1. $\gcd(k, 24) \in \{1, 2\}$**.** Take any code of length $12k$ and index ± 24 from [2]. For $m = 1$ take $g = 1$; for $m = 2$ take $g = -1$, which fixes no maximal codeword.

4.2. $\gcd(k, 24) = 4$**.** Label the Golay code G_{12} as in [2]. The signed permutation

$$J = (0\ 1)(2\ 3)(4\ 5)(6\ 7)(8\ 9)(10\ 11), \qquad \text{signs } (-,+,-,+,+,-,-,+,+,-,-,+),$$

meaning $J(e_i) = s_i e_{J(i)}$, lies in $\mathrm{Aut}(G_{12})$ and satisfies $J^2 = -1$ (it is one of 792 such elements of $\mathrm{Aut}(G_{12}) \cong 2.M_{12}$). The block $\beta = \{0, 1\} \sqcup \{2, 3\}$ is J-stable with J-stable pairs, and $R_\beta(G_{12}) = 24$. Apply Lemma 3.1 with $G = \langle J \rangle$. The two step-one blocks are J-stable; refine one of them, split into two J-stable pairs $\{p, Jp\} \sqcup \{p^*, (Jp)^*\}$. The result is a system of two J-stable blocks with J-stable pairs, and the construction iterates, adding 24 coordinates per step. By (1) and Lemma 3.1 we get, for every even k, a self-dual code of length $12k$ with Index $= \pm 24$ and $J \in \mathrm{Aut}$ with $J^2 = -1$. For $\gcd(k, 24) = 4$, J has order $4 = m$ and $J^{m/2} = -1$, so Theorem 1.1 applies.

4.3. $\gcd(k,24) \in \{3,6,12\}$**.** On 36 coordinates let g be the product of the six signed 6-cycles $e_{6j+i} \mapsto e_{6j+i+1}$ $(0 \le i < 5)$, $e_{6j+5} \mapsto -e_{6j}$. Then g has order 12, $g^6 = -1$, $J = g^3$ pairs the coordinates within each 6-cycle, and $h = g^4$ has order 3 with cycle shape 3^{12}. Following the search method of [6] (random g-invariant self-dual codes, keeping those with index ± 24 and $T_h = 0$), we found the code E_{36} of Appendix A, with $\mathrm{Index}(E_{36}) = 24$ and $T_h(E_{36}) = 0$; its collapsed fixed code D has length 12 and dimension 6.

A g-stable block is a union of two J-pairs from different 6-cycles; its h-orbit is a g-stable system of three blocks. For the systems

$$\sigma_3 = \{(24,27|30,33),(28,25|34,31),(26,29|32,35)\}$$

(a block $(a,b|c,d)$ is split as $\{a,b\} \sqcup \{c,d\}$) and

$$\sigma_6 = \sigma_3' \cup \{(12,15|25,28),(16,13|29,26),(14,17|27,24)\},$$
$$\sigma_3' = \{(0,3|7,10),(4,1|11,8),(2,5|9,6)\},$$

direct enumeration gives $R_{\sigma_3}(E_{36}) = R_{\sigma_6}(E_{36}) = -24$ and $R_{\bar\sigma_3}(D) = R_{\bar\sigma_6}(D) = 0$. Apply Lemma 3.1 with $G = \langle g \rangle$, refining at each step one h-orbit of J-stable step-one blocks with J-stable splits, and splitting each new block $\{x, y, x^{**}, y^{**}\}$ as $\{x,y\} \sqcup \{x^{**}, y^{**}\}$; this yields a g-stable system of six blocks and adds 72 coordinates. (With the split $\{x, x^{**}\} \sqcup \{y, y^{**}\}$ the stabilisers would drop to $\{\pm 1\}$ from the next step on, and the chains would skip residue classes.) Starting from σ_3 we reach every $k \equiv 0 \pmod 6$, from σ_6 every odd multiple of 3 other than 3, and $k = 3$ is E_{36} itself. By (1) and Lemmas 3.1 and 3.2, every code in these chains has Index $= \pm 24$, contains g with $g^6 = -1$ and all cycles of length 6, and has $T_h = 0$. Theorem 1.1 then applies with h for $m = 3$, with g^2 for $m = 6$ ($(g^2)^3 = -1$, $(g^2)^2 = h$), and with g for $m = 12$. □

5. Proof of Theorem B

The condition $g^4 = -1$ forces every coordinate cycle of g to have length 4 and an odd number of sign changes. For a maximal x we get $f(gx) = (-1)^{\#\text{cycles}} f(x) = (-1)^{3k} f(x)$, so for odd k the index vanishes. Let k be even. Then $f \circ g = f$, and $\langle g \rangle$ acts freely on nonzero vectors, since every nontrivial subgroup of $\langle g \rangle \cong \mathbb{Z}/8$ contains $g^4 = -1$. Hence $F(X) \in 8\mathbb{Z}$ for every g-invariant $X \subseteq \mathbb{F}_3^{12k} \setminus \{0\}$.

Structure. Over $\mathbb{F}_3$, $x^4 + 1 = (x^2 + x + 2)(x^2 + 2x + 2)$. On each 4-cycle the space $\mathbb{F}_3^4$ splits as $V_+ \oplus V_-$, the kernels of the two factors evaluated at g. Each is Lagrangian, and they are dual to each other. Hence g-invariant self-dual codes are exactly the codes $C = C_+ \oplus C_-$ with $C_+ \subseteq V_+^{3k} \cong \mathbb{F}_9^{3k}$ an arbitrary $\mathbb{F}_9$-subspace and C_- its annihilator in V_-^{3k}.

A four-code identity. J makes C an $\mathbb{F}_9$-space (J = multiplication by $\sqrt{-1}$), and the ternary form is the real part of a Hermitian form. Let $D \subseteq C$ be J-invariant of codimension 2. Then $D^\perp/D$ is a Hermitian hyperbolic plane over $\mathbb{F}_9$. Its 32 nonzero isotropic vectors form exactly four isotropic $\mathbb{F}_9$-lines, so exactly four J-invariant self-dual codes $C_1, \dots, C_4$ contain D. Every maximal vector is isotropic, and Poisson summation with $\hat f = 3^{n/2} f$ gives $F(D^\perp) = 9F(D)$. Hence

$$\mathrm{Index}(C_1) + \mathrm{Index}(C_2) + \mathrm{Index}(C_3) + \mathrm{Index}(C_4) = 12\,F(D). \tag{2}$$

Induction. Let $D_+ \subset C_+$ be an $\mathbb{F}_9$-hyperplane and $D = D_+ \oplus C_-$. Then $D^\perp/D = L_+ \oplus L_-$ is the sum of the two g-eigenlines, so $C_1 = C$ and $C_2 = D_+ \oplus D_+^\circ$ are the g-invariant codes through D (D_+° the annihilator of D_+ in V_-^{3k}), while g, which acts on $L_\pm$ by the two square roots $\pm\lambda \in \mathbb{F}_9$ of $J = \sqrt{-1}$, maps the isotropic line $\mathbb{F}_9(u+u')$ ($u \in L_+$, $u' \in L_-$) to the isotropic line $\mathbb{F}_9(u-u')$ and so swaps C_3 and C_4. As g preserves f, $\mathrm{Index}(C_3) = \mathrm{Index}(C_4) \in 24\mathbb{Z}$ [6, 2]; and $F(D) \in 8\mathbb{Z}$. Now (2) gives $\mathrm{Index}(C) + \mathrm{Index}(C_2) \equiv 0 \pmod{48}$. Removing one dimension

of C_+ at a time reaches $C_+ = 0$, i.e. $C = V_-^{3k}$, a direct sum of tetracodes with index 0. Hence $48 \mid \mathrm{Index}(C)$. □

6. The quaternion group

Theorem B concerns cyclic groups. The quaternion group $Q_8 = \langle J, K\rangle$, with $J^2 = K^2 = -1$ and $JK = -KJ$ in $\mathrm{Aut}(C)$, has no element of order 8. In Q_8 every commuting pair generates a cyclic group, and every nontrivial one contains -1, which fixes no maximal codeword. So the twisted-sector argument of [6] gives index $\mathrm{Index}(C)/8$ for $V_{\Lambda(C)}/\!\!/Q_8$, *provided the Q_8-action is non-anomalous.* We write $\mathcal{A}(G) = \mathrm{Hom}(\Omega_3^{\mathrm{Spin}}(BG), U(1))$ for the group of anomalies of G-actions on two-dimensional fermionic theories; it is the extended supercohomology group in which [6, Thm. 4.2] places the anomaly [8, §2.4]. We use the following standard fact.

Lemma 6.1 (Coprime gauging)**.** *Let $\Gamma = A \rtimes K$ act by even automorphisms on a holomorphic SVOA V, where A is abelian and $\gcd(|A|, |K|) = 1$. Suppose the anomaly of Γ is $p^*\varepsilon$, where $p\colon \Gamma \to K$ is the projection and $\varepsilon \in \mathcal{A}(K)$ has vanishing Majorana layer. Let $V/\!\!/A$ be an orbifold by A with K-invariant discrete torsion. Then $\hat{A} \rtimes K$ acts on $V/\!\!/A$ with anomaly $p^*\varepsilon$, and K acts on $(V/\!\!/A)^{\hat{A}} = V^A$ as before. In particular $K \subseteq \hat{A} \rtimes K$ acts on $V/\!\!/A$ with anomaly ε.*

Sources. For pointed fusion categories: the orbifold with trivial discrete torsion corresponds to the module category $\mathcal{M}(A, 1)$ over $\mathrm{Vec}(\Gamma, p^*\varepsilon)$. Its dual category is $\mathrm{Vec}(\hat{A} \rtimes K, p^*\varepsilon)$ by [10, Thm. 4.9], explicitly [12, Thm. 3.7] with $F = \hat{F} = 0$; the dual extension splits since $H^2(K; \hat{A}) = 0$. A K-invariant discrete torsion $\mu \in H^2(A; U(1))^K = H^2(\Gamma; U(1))_{(|A|)}$ extends to a 2-cocycle $\tilde{\mu}$ on Γ, and twisting the tensor structure by $\tilde{\mu}$ is an autoequivalence of $\mathrm{Vec}(\Gamma, p^*\varepsilon)$ that carries $\mathcal{M}(A, 1)$ to $\mathcal{M}(A, \mu)$. The passage to VOAs identifies holomorphic extensions of V^Γ with Lagrangian algebras in $\mathrm{Rep}(V^\Gamma) \simeq Z(\mathrm{Vec}(\Gamma, p^*\varepsilon))$, i.e. with module categories [8, Prop. 2.2.1 and the proof of Prop. 2.3.1]. For bosonic VOAs the remaining input is the regularity of V^Γ, known for solvable Γ [4]. The super case reduces to the bosonic one through bosonic shadows, which exist because the Majorana layer vanishes [8, §2.4], [3]; this is at the level of rigour of [6, Thm. 4.2]. In physical terms the lemma is [11, §2.4, Claim] with trivial extension classes. □

Theorem 6.2. *Let C be self-dual of length $12k$ and $Q_8 = \langle J, K\rangle \subseteq \mathrm{Aut}(C)$ with $J^2 = K^2 = -1$. The anomaly of the Q_8-action on $V_{\Lambda(C)}$ is $3k \cdot a$, where $a \in \mathcal{A}(Q_8)$ is the anomaly of Q_8 acting on eight Majorana fermions in its real regular representation, and a has order 8. In particular the action is non-anomalous if and only if $8 \mid k$.*

Proof. Since $J^2 = -1$, J fixes no coordinate, and likewise K and JK; so $Q_8/\{\pm 1\}$ acts freely on coordinates, with $3k$ orbits of size 4. In the coordinates of Lemma 6.5 the Q_8-invariant tetracodes on an orbit are $\mathbb{F}_3^2 \otimes \ell$ for the four lines $\ell \subseteq \mathbb{F}_3^2$; choose one on each orbit and let $C_0 = \mathbb{F}_3^2 \otimes U_0$ be their direct sum.

Reduction to C_0. Write $C = \mathbb{F}_3^2 \otimes U$ (Lemma 6.5) and put $D = C \cap C_0 = \mathbb{F}_3^2 \otimes (U \cap U_0)$ and $W = \mathbb{F}_3^2 \otimes X$ for a complement X of $U \cap U_0$ in U_0; W is Q_8-stable. Since $\langle X, Y\rangle = \Omega(x_2, y_1) - \Omega(x_1, y_2)$ for rows x_i, y_i, we have $(\mathbb{F}_3^2 \otimes X)^\perp = \mathbb{F}_3^2 \otimes X^\Omega$. Hence $W \cap C = 0$; $C \cap W^\perp = \mathbb{F}_3^2 \otimes (U \cap X^\Omega) = D$, because an element of $U \cap X^\Omega$ is Ω-orthogonal to $U_0 = X + (U \cap U_0)$ and so lies in $U_0^\Omega = U_0$; and $C_0 = D \oplus W$ with W totally isotropic. For $w \in W$ let $h_w = \tilde{w}/\sqrt{3}$ with $\tilde{w} \in \{0, \pm 1\}^{12k}$. Then $\exp(2\pi i\, h_w(0))$ multiplies the frame module $V_{\sqrt{3}\mathbb{Z}^{12k} + \tilde{c}/\sqrt{3}}$, $c \in C$, by $e^{2\pi i\, w \cdot c/3}$. These automorphisms form a group $\hat{W} \cong W$, faithful since $W \to \mathrm{Hom}(C, \mathbb{F}_3)$ is injective, and Q_8 normalises it. Its fixed points are $V_{\Lambda(C \cap W^\perp)} = V_{\Lambda(D)}$, the invariant parts of its twisted sectors are the $V_{\Lambda(D) + h_w}$, and these add up to $V_{\Lambda(D \oplus W)} = V_{\Lambda(C_0)}$. So $V_{\Lambda(C_0)} = V_{\Lambda(C)}/\!\!/\hat{W}$. In particular $\hat{W}$ is non-anomalous, and the discrete torsion of this orbifold is Q_8-invariant because C_0 is.

Let $\Gamma = \hat{W} \rtimes Q_8$ with projection $p\colon \Gamma \to Q_8$, and let α be the anomaly of Q_8 on $V_{\Lambda(C)}$. Since $\hat{W}$ has odd order, $BQ_8 \to B\Gamma$ is a $\mathbb{Z}_{(2)}$-homology equivalence, so the 2-primary part of the anomaly of Γ is $p^*\alpha$. Its 3-primary part restricts injectively to $\hat{W}$ (corestriction after transfer is multiplication by 8), where it vanishes. So the anomaly of Γ is $p^*\alpha$. The Majorana layer of α vanishes: J and K are products of $6k$ transpositions with sign product -1, so $\langle \chi, (1-g)\chi\rangle = 36k \in 4\mathbb{Z}$ for $\chi = \sqrt{3}(1,\dots,1)$ [6, Ex. 4.4], and the layer is a homomorphism $Q_8 \to \mathbb{Z}/2$. By Lemma 6.1 a complement $Q_8' \cong Q_8$ in $\hat{W}^\vee \rtimes Q_8$ acts on $V_{\Lambda(C_0)}$ with anomaly α, and on $V_{\Lambda(D)}$ it acts as Q_8 does. In particular it acts on the frame $V_{\sqrt{3}\mathbb{Z}}^{\otimes 12k} \subseteq V_{\Lambda(D)}$ as the canonical lift of [6, Thm. 3.1] for C_0 does. The automorphisms of $V_{\Lambda(C_0)}$ that are trivial on the frame form the 3-group $\mathrm{Hom}(C_0, \mathbb{C}^\times)$ (Schur's lemma on the frame modules). So the two actions differ by a 1-cocycle with values in a 3-group, hence are conjugate, and the canonical Q_8-action on $V_{\Lambda(C_0)}$ has anomaly α.

Free fermions. The lattice of a tetracode is odd unimodular of rank 4, hence $\mathbb{Z}^4$, so $V_{\Lambda(C_0)} \cong \bigotimes_{3k} V_{\mathbb{Z}^4} \cong \bigotimes_{3k} \mathrm{Fer}(8)$ with Q_8 acting factorwise. An automorphism of Fer(8) is determined by its action on the weight-$\frac{1}{2}$ space, which is spanned by the $e^{t/\sqrt{3}}$, t a nonzero tetracode word. Q_8 permutes these up to sign and freely, so it acts by the regular character: the weight-$\frac{1}{2}$ space is the real regular representation $\mathrm{Reg} = 1 \oplus \chi_i \oplus \chi_j \oplus \chi_k \oplus \mathbb{H}$. Hence the anomaly is $3k \cdot a$ with a the anomaly of Fer(Reg).

$8a = 0$. Anomalies add under tensor products. A Majorana fermion on which Q_8 acts by a sign character χ contributes $\chi^*\nu$, with ν the generator of $\mathcal{A}(\mathbb{Z}/2) \cong \mathbb{Z}/8$. The four fermions in $\mathbb{H}$ are acted on through $Q_8 \subset SU(2)_L \subset SO(4)$, and $SU(2)_L$ acts on four Majorana fermions at level one; this anomaly is the class $c_2 \in H^4(BSU(2);\mathbb{Z})$, which pulls back into $H^4(BQ_8;\mathbb{Z}) \cong \mathbb{Z}/8$. So $8a = 0$.

a has order 8. On a spherical space form $M = S^3/\Gamma$, $\Gamma \subseteq SU(2)$, with spin structure $\varepsilon\colon \Gamma \to \{\pm 1\}$, the anomaly of fermions in a real representation ρ is $\frac{1}{2}(\eta(\rho) - \dim \rho\, \eta(1)) \bmod 1$, where $\eta(\rho) = -\frac{1}{|\Gamma|}\sum_{g\neq 1} \varepsilon(g)\chi_\rho(g)/(2 - \mathrm{tr}\, g)$ [5]. (One Majorana fermion odd under $\mathbb{Z}/2$ gives $\pm\frac{1}{8}$ on $\mathbb{RP}^3$, the generator.) For $\rho = \mathrm{Reg}$ this gives $\frac{5}{8}$ on S^3/Q_8 for all four spin structures, $\frac{1}{4}$ on the lens spaces $S^3/\mathbb{Z}_4$, and $\frac{1}{2}$ on $\mathbb{RP}^3$. So a has order 8, and $3k \cdot a = 0$ forces $8 \mid k$. □

Remark 6.3. On cyclic subgroups Theorem 6.2 can be checked without the reduction step. For $g \in \{J, K, JK\}$ the Frame shape is $2^{-6k}4^{6k}$, and [6, Ex. 4.4] gives the anomaly $2k \in \mathbb{Z}/8$ of $\langle g\rangle$ on $V_{\Lambda(C)}$ directly; for $g = -1$ it is $1^{-12k}2^{12k}$ and the anomaly is $2k \in \mathbb{Z}/4$. For free fermions a rotation plane of angle $2\pi q/m$ contributes q^2. So Fer(Reg) restricted to $\langle i\rangle$ (two planes of angle $\pi/2$ and one of angle π) gives 6, restricted to $\langle -1\rangle$ (two planes of angle π) it gives 2, and indeed $3k \cdot 6 \equiv 2k \pmod 8$ and $3k \cdot 2 \equiv 2k \pmod 4$. The eta invariants on the lens spaces and $\mathbb{RP}^3$ agree as well, and reproduce the anomaly clause of Theorem 1.1: on $\mathbb{Z}/4$ and $\mathbb{Z}/2$ the anomaly vanishes exactly when $4 \mid k$ and $2 \mid k$. This check cannot see $4a$. That class restricts to zero on every cyclic subgroup, and every abelian subgroup of Q_8 is cyclic, but it is nonzero on S^3/Q_8.

The class $4a$ can be seen on $V_{\Lambda(C)}$ itself, through the (-1)-twisted sector. Each $V_{\sqrt{3}\mathbb{Z}}$ has a θ-twisted module with one odd zero mode γ_i. The θ-twisted modules of the frame are completely reducible [1, Thm. 4.2], and the θ-twisted $V_{\Lambda(C)}$-module and the irreducible frame module both have lowest-weight spaces of dimension $2^{n/2}$ and the same character. So the θ-twisted $V_{\Lambda(C)}$-module is the unique irreducible θ-twisted module of the frame, with ground states the spinor module T of the Clifford algebra $\langle \gamma_1, \dots, \gamma_{12k}\rangle$. By Schur's lemma, Q_8 therefore acts on it through $Q_8 \subseteq SO(12k) \to \mathrm{Pin}(12k)$. In particular this action does not depend on C, and $T \cong (\mathbb{H} \oplus 2\psi)^{\otimes 3k}$, with $\mathrm{Tr}_T(JK)/(\mathrm{Tr}_T J \cdot \mathrm{Tr}_T K) = 2^{-3k}$. Now normalise the lifts of J, K and JK on the (-1)-twisted sector by modular covariance: apply ST^2S to the twining character

$\eta(2\tau)^{6k}/\eta(4\tau)^{6k}$. For $4 \mid k$ these lifts are the canonical ones of the non-anomalous cyclic subgroups, and their trace on T is $(-1)^{k/4}2^{3k}$. They satisfy $\hat{J}\hat{K} = \mu\,\widehat{JK}$ with $\mu = (-1)^{k/4}$. If Q_8 were non-anomalous, they would form a representation of Q_8, i.e. $\mu = 1$. To see this, suppose Q_8 is non-anomalous and let $W = V_{\Lambda(C)}/\!\!/Q_8 = \bigoplus_{[g]} V(g)^{C(g)}$. The characters of its sectors are explicit. The (-1)-sector contributes $\frac{1}{8}E + \frac{1}{4}\big(\sum_x \psi(x)t_x\big)F$, for some character ψ of Q_8, where $t_x = 2^{3k}$ and $F = q^{n/48}\prod_{r\ \mathrm{odd}}(1+q^r)^{-n/2}$. Each class $\{\pm x\}$ contributes $\frac{1}{4}\big(g(\tau)+g(\tau+2)\big)$: write $\rho_x(x) = \varepsilon e^{2\pi i L_0}P$. On the weight space $h_x + j/4$ odd powers of x have trace 0 and x^2 acts by $\varepsilon^2 e^{4\pi i h_x}(-1)^j$, so the invariants are nonzero at $j=0$ or $j=1$. There $\rho_x(x) = 1$ while $e^{2\pi i L_0} = \pm P$ (spin–statistics in W), so $\varepsilon = \pm 1$ and x^2 acts as $e^{4\pi i L_0}$. The untwisted sector contributes $\frac{1}{8}\big(Z(1,1) + e + 6f\big)$. Here $e = \eta(\tau)^{12k}/\eta(2\tau)^{12k}$, $f = \eta(2\tau)^{6k}/\eta(4\tau)^{6k}$, and E, g are the characters of the (-1)- and x-twisted modules. The exact identities $e|S = E$, $f|S = g$ and $f|ST^2S = N\,F$ with $N = (\sqrt{2}\,e^{2\pi i/48})^{6k} = 2^{3k}e^{2\pi i k/8}$ hold (in the last one the two eta multipliers cancel). So $\operatorname{ch} W|S - \operatorname{ch} W = \frac{1}{4}\big(3N - \sum_x \psi(x)t_x\big)\big(F - g(\tau+2)/N\big)$. The NS character of the holomorphic SVOA W is S-invariant, so $\psi(J)+\psi(K)+\psi(JK) = 3e^{2\pi i k/8}$, which forces $8 \mid k$. So for $k \equiv 4 \pmod 8$ the Q_8-action is anomalous, independently of the reduction step. (For every k the same computation gives $\mu = e^{-2\pi i k/8} = \exp(2\pi i \cdot 3k \cdot \frac{5}{8})$, the value of $3k\cdot a$ on S^3/Q_8.)

This also gives a second proof of Theorem 6.2, without the reduction step. The anomaly of Q_8 lies in restricted supercohomology $r\mathrm{SH}^3(Q_8)$, since its Majorana layer vanishes, and this group has order at most $|H^3(Q_8;U(1))| \cdot |H^2(Q_8;\mathbb{Z}/2)| = 32$. Free fermions $\mathrm{Fer}(\rho)$ with $w_1(\rho) = 0$ realise 32 classes that are distinguished by their restrictions to the three cyclic subgroups of order 4 together with their eta invariants on S^3/Q_8. So only 0 and $\kappa := 4a$ restrict to zero on all cyclic subgroups. For $4 \mid k$ the anomaly of $V_{\Lambda(C)}$ has vanishing cyclic restrictions, so it lies in $\{0,\kappa\}$; it equals κ when $k \equiv 4 \pmod 8$, because $\mu = -1$. For $8 \mid k$, pick any Q_8-invariant code C' with $k' \equiv 4 \pmod 8$. Then $V_{\Lambda(C\oplus C')} = V_{\Lambda(C)} \otimes V_{\Lambda(C')}$ and anomalies add. Both $C \oplus C'$ and C' have anomaly κ, so the anomaly for C is $\kappa - \kappa = 0$. Adding sums of tetracodes, whose anomaly is $3k\cdot a$ by the free-fermion step, extends this to $3k\cdot a$ for every k. The values are also consistent with topological modular forms: when $4 \mid k$ but $8 \nmid k$, a non-anomalous Q_8 would produce index 3, which is not divisible by $24/\gcd(k,24) \in \{6,2\}$.

Corollary 6.4. *For every $k \equiv 0 \pmod 8$ there is a holomorphic $\mathcal{N}=1$ SVOA of central charge $12k$ and index 3. In particular the Main Question has a positive answer for $\gcd(k,24) = 8$.*

Proof. Random Q_8-invariant self-dual codes of length 24 reach index ± 24 (e.g. 27 of 100 samples). From such a code with a Q_8-stable two-block system of value $R = 24$, Lemma 3.1 with $G = Q_8$ gives Q_8-invariant codes of index ± 24 at every $k \equiv 0 \pmod 4$ (checked explicitly up to length 96). For $8 \mid k$, Theorem 6.2 makes the action non-anomalous, so the orbifold exists [6, Thm. 4.2]; it is $\mathcal{N}=1$ because Q_8 fixes the supercurrent, and its index is $24/8 = 3$. Indeed, by supersymmetry the index counts Ramond ground states (weight $c/24$). Every $g \in Q_8 \setminus \{1\}$ has even order m, a fixed-point-free coordinate permutation unless $g = -1$, and $g^{m/2} = -1$; so its eigenvalues on $\mathbb{R}^n$ are the odd powers of $e^{2\pi i/m}$, each $2n/m$ times, and its twisted Ramond sector has lowest weight $\frac{n}{24} + \frac{n}{12m^2} > \frac{c}{24}$. In the untwisted sector the ground states are the maximal codewords, and no element of even order fixes one ($g^{m/2} = -1$ would fix it, but $-w \neq w$). So only the Q_8-average of the untwisted ground states contributes, giving $\mathrm{Index}(C)/8 = 3$ (the Q_8 seed has $R = +24$, so the chain codes have Index $= +24$). Here we use the Ramond decomposition $W_R = \bigoplus_{[g]} V_{R,g}^{C(g)}$ of the orbifold, as in the proof of [6, Thm. 5.6]; like the fermionic case of [6, Thm. 4.2] itself, its super-VOA proof is only sketched there. The same argument, with [6]'s modular step for the order-3 terms, gives the index of the $Q_8 \times \mathbb{Z}/3$ orbifold below. □

For $k \equiv 0 \pmod{24}$ the same reasoning applies to $Q_8 \times \mathbb{Z}/3$, with the order-3 element h fixed-point-free and balanced: the anomaly vanishes (Theorem 6.2, [6] for the $\mathbb{Z}/3$ part, coprimality for mixed terms), and the orbifold has index 1. A suitable code is constructed in §6.2.

6.1. **Morita form.** The following description makes Q_8-invariant codes computable well beyond direct enumeration. Let $A = \mathbb{F}_3\langle i, j\rangle$ be the quaternion algebra over $\mathbb{F}_3$, identified with $M_2(\mathbb{F}_3)$ by $i \mapsto \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $j \mapsto \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$.

Lemma 6.5. *Let $Q_8 = \langle J, K\rangle$ act on $\mathbb{F}_3^{4r}$ by signed permutations with $J^2 = K^2 = -1$. After a monomial change of coordinates each of the r orbits is A with the coordinates of $a+bi+cj+dk$, Q_8 acts by left multiplication, and the dot product is the polar form of the norm* det. *Consequently the Q_8-invariant codes are exactly*

$$C = \mathbb{F}_3^2 \otimes U = \{X \in M_{2\times 2r}(\mathbb{F}_3) : \textit{both rows of } X \textit{ lie in } U\}, \qquad U \subseteq \mathbb{F}_3^{2r},$$

and C is self-dual if and only if U is Lagrangian for $\Omega = \bigoplus_b \omega_b$, $\omega(x, y) = x_0y_1 - x_1y_0$. A block X_b has weight 4 if and only if $X_b \in SL_2(\mathbb{F}_3) \setminus Q_8$, and then $f(X_b) = \varphi(\psi(X_b))$, where $\psi\colon SL_2(\mathbb{F}_3) \to \mathbb{Z}/3$ is the abelianisation.

Proof. The orbit module is induced from the sign character of $\{\pm 1\}$, i.e. $\mathbb{F}_3[Q_8]/(1 + z) \cong A$; submodules of A^r are $\mathbb{F}_3^2 \otimes U$ by Morita equivalence, and on them the polar form of the norm is $\langle X, Y\rangle = -(\Omega(x_1, y_2) - \Omega(x_2, y_1))$. The norm of an element of $\mathbb{F}_3^4$ is its weight mod 3; the norm-one elements form $SL_2(\mathbb{F}_3) = Q_8 \cup \{(\pm 1 \pm i \pm j \pm k)/2\}$ with $1/2 = -1$, and f is Q_8-bi-invariant with opposite signs on the two nontrivial cosets. □

Left multiplication by $SL_2(\mathbb{F}_3)$ preserves C, acts freely on the codewords all of whose blocks lie in $SL_2(\mathbb{F}_3)$, and shifts every $\psi(X_b)$ by the same amount. Hence

$$\operatorname{Index}(C) = 8 \sum_{\Pi} \sum_{s\in\mathbb{F}_3} \prod_b \varphi(\psi_b(\Pi) + s),$$

the sum running over these orbits, each represented by its unique element with $X_{b_0} = 1$. The same holds for R_σ when σ consists of Q_8-orbits, with the block weight equal to $(-2, 1, 1)$ as a function of ψ. Enumerating these representatives costs 3^{r-2} linear solves; at length 72 this takes seconds, whereas C has 3^{36} words.

With h added, h-invariant Lagrangians are submodules over $R = \mathbb{F}_3[\mathbb{Z}/3] = \mathbb{F}_3[u]/u^3$. The free ones are $U = R \cdot \{e_i + u(A_1 f)_i + u^2((A_1 + S)f)_i\}$ over a Lagrangian $\bar{U} = \langle e\rangle$ of $\mathbb{F}_3^{2k}$, with A_1 skew and S symmetric, and then $T_h = \operatorname{Index}(\mathbb{F}_3^2 \otimes \bar{U})$. For non-free U, Poisson summation gives $72 \mid T_h$. At length 36 we found (by exhaustive and random checks at lengths 12, 24, 36) the exact law $\operatorname{Index} \equiv T_h \cdot A_1(P)^2 \pmod{72}$ for free U, where $P \subseteq \bar{U}$ is spanned by the three lines of $\bar{U}$ vanishing on one block. So at length 36 a vanishing twined count forces $72 \mid \operatorname{Index}$, and equivariant chains seeded there cannot reach index ± 24 with $T_h = 0$. *This is special to length* 36: at length 72 there are $Q_8 \times \mathbb{Z}/3$-invariant self-dual codes with $T_h = 0$ and $\operatorname{Index} = -3840 \not\equiv 0 \pmod{72}$, and one with $T_h = 0$ and $\operatorname{Index} = 24$. A chain seeded at length 72 with a G-stable system of six orbit blocks reaches every $k \equiv 0 \pmod{24}$ with $\operatorname{Index} = R_\sigma$ and $T_h = R_{\bar\sigma}(D)$ (Lemma 3.2).

6.2. **Proof of Theorem D.** A random search over free lifts of Lagrangians $\bar{U}$ whose collapsed code has $R_{\bar\sigma} = 0$ for all fifteen systems of two h-triples found, after about 7000 systems, a Lagrangian $U \subseteq \mathbb{F}_3^{36}$ (free over R) such that $E = \mathbb{F}_3^2 \otimes U$ satisfies

$$R_\sigma(E) = 24, \qquad R_{\bar\sigma}(D) = 0, \qquad \operatorname{Index}(E) = -1608, \qquad T_h(E) = 0,$$

for σ the six orbit blocks of two h-triples. Here $R_\sigma(E)$ was computed by the good-plane enumeration with two different normalisations and by a second implementation, and $R_{\bar\sigma}(D)$ by

enumerating the length-24 code D. Lemma 3.1 with $G = Q_8 \times \langle h \rangle$, refining the G-orbit of one step-1 block at each step, gives G-invariant self-dual codes of lengths $144, 288, 432, \ldots$, i.e. every $k \equiv 0 \pmod{12}$ with $k \geq 12$. Their index is $R_\sigma(E) = 24$ [2, Lemma 5.8] and their twined count is $R_{\bar{\sigma}}(D) = 0$ (Lemma 3.2); $J^2 = K^2 = -1$ and the cycle shape 3^{4k} of h are inherited. For $24 \mid k$ the G-action is non-anomalous: the Q_8 part by Theorem 6.2, the $\mathbb{Z}/3$ part by [6], and mixed terms vanish since the orders are coprime. Every abelian subgroup of G is cyclic. Those containing -1 have vanishing twisted-twined indices, and the only other nontrivial one is $\langle h \rangle$, with $T_h = 0$. So the orbifold has index $24/24 = 1$, and it is $\mathcal{N} = 1$ since G fixes the supercurrent. □

Remark 6.6 (Verification)**.** All statements were checked by computer. The chains of §4.2 and §4.3 were built explicitly up to lengths 144 and 108: invariance, $g^6 = -1$ (resp. $J^2 = -1$), cycle shapes and self-duality at every step; $|\,\mathrm{Index}\,| = 24$ at lengths 48 and 72; and $T_h = 0$ by enumerating the fixed codes at lengths 72, 108 and 144. Identity (2) was checked sign-exactly on random configurations at length 24, the induction step at lengths 24 and 48, and $48 \mid \mathrm{Index}$ on random codes with $g^4 = -1$ at lengths 24 to 72. Automorphism groups were computed with nauty/Traces [9]: the index-24 codes of [2] of lengths 36 and 72 have automorphism groups of 2-power order, so they do not satisfy condition (2); the one of length 48 does satisfy Theorem 1.1 with an element J, $J^2 = -1$. For Theorem 6.2: on the Q_8-invariant codes of lengths 24, 48 and 96 every coordinate orbit carries an invariant tetracode and their sum is a Q_8-invariant self-dual code; the identities $W \cap C = 0$, $C \cap W^\perp = D$ and $C_0 = D \oplus W$ of the reduction step hold on these codes and on random Q_8-invariant codes up to length 96; the eta invariants and the cyclic anomalies of Remark 6.3 (for $k \leq 48$) were evaluated in exact rational arithmetic. The Pin lifts on the frame Clifford module were checked on the Q_8-invariant codes of lengths 24, 48, 96, and the modular normalisation was evaluated numerically, agreeing with $2^{3k} e^{2\pi i k/8}$ to 10^{-14}. The count of 32 free-fermion classes, and the direct-sum argument on the codes of lengths 48, 96 and 144, were checked by computer. The good-plane enumeration of §6.1 agrees sign-exactly with direct enumeration at lengths 12, 24, 36, and in absolute value with an independent index algorithm at length 72. For Theorem D the chain was built explicitly to lengths 144 and 288: self-duality, invariance under J, K, h, the relations, and the cycle shape of h at both lengths, and $T_h = 0$ at length 144 by computing the index of the collapsed fixed code. Code and data are available at [repository].

Appendix A. The seed code E_{36}

Generator matrix of E_{36} (coordinates $0, \ldots, 35$, entries in $\{0, 1, 2\}$):

```
100000000000000000010120122110112102
010000000000000000021012022211011210
001000000000000000021012022211011121
000100000000000000102101202221201112
000010000000000000210210220222120111
000001000000000000210211220222120111
000000100000000000101222200220000211
000000010000000000110122020022200021
000000001000000000111012102002220002
000000000100000000111101110200122000
000000000010000000211110011020012200
000000000001000000021111001102001220
000000000000100000112012221112122212
000000000000010000111201122111112221
000000000000001000211120212211211222
000000000000000100021112221221121122
000000000000000010102111222122112112
```

`00000000000000000121021112221211121 1`

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